This account should be able to run the *sudo* commands without a root password.

Since two provider drivers for load balancers are being compared, two setups are used to install OpenStack — OVN is installed in one setup and the default configuration is used in the other one. Horizon (OpenStack dashboard) and Cinder (block storage) services are disabled in both the setups to improve performance and increase the amount of free resources, since the current objective is not dependent on either of the services.

Creating the stack user: Run the following command to create the stack user. Here, stack is the user name of the user:

```
sudo adduser stack
```

Hit *Enter* for all other fields after retyping the password.

Octavia.create_and_list_loadbalancers (62.244s)

Create a loadbalancer per each subnet and then list loadbalancers.

Overview Details Failures Input task

Load duration: **24.974 s** Full duration: **62.244 s** Iterations: **5** Failures: **5** Started at: **2020-11-20T07:23:24**

Service-level agreement

Criterion	Detail	Success
failure_rate	Failure rate criteria 0.00% <= 100.00% <= 0.00% - Failed	False

Total durations

Action	Min (sec)	Median (sec)	90%ile (sec)	95%ile (sec)	Max (sec)	Avg (sec)	Success	Count
octavia.load_balancer_create	1.35	2.185	2.854	3.071	3.288	2.204	100.0%	5
octavia.wait_for_loadbalancers	6.182	6.217	8.274	8.277	8.28	7.025	0.0%	5
-> octavia.load_balancer_show (x4)	*0.177*	*0.178*	*0.205*	*0.208*	*0.212*	*0.189*	*100.0%*	*3*
-> octavia.load_balancer_show (x5)	*0.26*	*0.264*	*0.268*	*0.268*	*0.269*	*0.264*	*100.0%*	*2*
total	**8.212**	**8.387**	**10.784**	**11.168**	**11.553**	**9.23**	**0.0%**	**5**
-> duration	*8.212*	*8.387*	*10.784*	*11.168*	*11.553*	*9.23*	*0.0%*	*5*
-> idle_duration	*0*	*0*	*0*	*0*	*0*	*0*	*0.0%*	*5*

Figure 3: Result of *Octavia.create_and_list_loadbalancers* task using Amphora

Octavia.create_and_list_loadbalancers (288.678s)

Create a loadbalancer per each subnet and then list loadbalancers.

Overview Details Input task

Load duration: **164.194 s** Full duration: **288.678 s** Iterations: **5** Failures: **0** Started at: **2020-11-18T06:23:22**

Service-level agreement

Criterion	Detail	Success
failure_rate	Failure rate criteria 0.00% <= 0.00% <= 0.00% - Passed	True

Total durations

Action	Min (sec)	Median (sec)	90%ile (sec)	95%ile (sec)	Max (sec)	Avg (sec)	Success	Count
octavia.load_balancer_create	2.02	3.946	123.48	123.739	123.998	51.204	100.0%	5
octavia.wait_for_loadbalancers	0.038	0.064	35.234	35.379	35.524	14.095	100.0%	5
-> octavia.load_balancer_show (x14)	*8.202*	*8.345*	*8.46*	*8.474*	*8.488*	*8.345*	*100.0%*	*2*
-> octavia.load_balancer_show	*0.038*	*0.048*	*0.061*	*0.063*	*0.064*	*0.05*	*100.0%*	*3*
octavia.load_balancer_list	0.058	0.179	0.288	0.307	0.326	0.177	100.0%	5
total	**2.117**	**4.104**	**158.908**	**159.058**	**159.208**	**65.494**	**100.0%**	**5**
-> duration	*2.117*	*4.104*	*158.908*	*159.058*	*159.208*	*65.494*	*100.0%*	*5*

Figure 4: Result of *Octavia.create_and_list_loadbalancers* task using OVN

Now, log in to the root user:

```
sudo -i
```

Give permissions to run *sudo* commands without a root password. Care should be taken while executing the line below, because mistakes made while executing the command can result in syntax errors or loss of configuration for other users. This can result in not letting other users use *sudo* commands. So, take a backup of the *sudoers* file before making any changes to it.

```
sudo cp /etc/sudoers /etc/sudoers.
original
```

Now execute the command given below:

```
echo "stack ALL=(ALL) NOPASSWD: ALL" >>
/etc/sudoers
```

The *exit* command is used to exit from the root and current user.

DevStack installation: First, enable the universe repository as DevStack installs packages and configure OpenStack:

```
sudo apt-add-repository universe
```

Upgrade the operating system by installing the latest updates and install *git* (to clone repositories), *bridge-utils* (to configure Octavia) and *python-is-python3* (for setting default Python version).

```
sudo apt update && sudo apt upgrade
sudo apt install git bridge-utils
python-is-python3
```

Clone the DevStack repository from OpenDev:

```
git clone https://opendev.org/
openstack/devstack
```

Enter the *devstack* directory (*cd devstack*).